

From the 2025/2026 school year

MS abbreviates “middle school” SM abbreviates “Stand-up Maths” MP is Meru Pras-tarah, also known as the Pascale Triangle SG is the Serpienski Gasket

From Idea To Calculation To Code to Experiment to Project

Some of these resonate with material in the “Puzzles” section.

- Ms. Halfway
 - Careful use of three markers: initialize $h = 0$, $m_0 = 0$, $m_1 = 1$
 - 1/2-arrival is the average of m_0 and m_1 (teach as an aside, then apply)
 - The last aspect is getting the update order correct
 - h should be the average of m_0 and m_1
 - m_1 should be updated before m_0 : Why?
 - m_0 will be updated to be h : $m_0 = h$
 - The concept of m_1 actually depends on (new) h and (old) m_0 ...
 - ...so calculate m_1 before changing m_0
 - Where does Ms. 2/7 wind up? Ms. a/b ?
 - Leads to Chaos Game
 - With an interesting underlying explanation
 - Advanced coding: Roll all 3 possibilities (All The Myriad Ways)
 - Hausdorf dimension
 - Draw the XOR universe (turtle)
 - This is probably easier to do starting with 3 cells > 1 cell
 - It avoids the jiggle back and forth finesse; maybe saves time
 - requests
 - music21

Coding Mechanics

Coding unto itself

- Execution flow is top down; but sometimes this rule breaks
 - Loops break this
 - Conditional logic breaks this
 - Functions are read but not run until called
 - Indented code blocks are called just that: *blocks* or (formally) a *suite*.
- Indentation syntax ~ multiple lines as a group activity
 - This suggests functions
- The turtle library cheat sheet
 - `turtle.Screen`
 - `tracer(0, 0)` and `update()`
 - `turtle.Turtle()`
 - `t.towards(x, y)` and `t[0].towards(t[1])`
 - `t.setheading(direction)` and `composition (randint(etc))`
 - `t.dot(25)`, `t.penup()`, `t.pendown()`
 - `t.pencolor("blue")`
 - `turtle.done()` is the means of avoiding blip-draw-gone syndrome

Short exercises that emphasize code mechanics and syntax

- Input gives a string but we want an integer
- Remainder function %
- Is a number even?
- Is a number divisible by n?
- Is a number prime?
- What is n!
- Alternating sign sequence
- Printing a triangle where the start location moves backwards
- Use sleep to instill some interactivity interest
- Use time to see how long something took

Puzzles and enlightening solutions

- Look and Say
 - “1... What is the next number?”
- Duck and Wolf
 - For MS the inner radius calculation tends to be too much
 - To this end an alternative is to match the shadow to the wolf...
 - ...where this gives us $r/4$ from the center
 - ...and then we can move “one feather towards the center”
 - ...now the shadow moves faster than the wolf
 - ...so it becomes a bit Argument by Enthusiasm
- Boring Worm
- Poker chip
- Gold Key
- Flipping locks
- 4 Hats and other prisoner/hat variants
- What is the graph of the state space of a hexahexaflexagon?
- Cellular automata
 - Begin in one dimension plus time
 - The AND, OR and XOR Universes
 - For a bit more fun make the universe closed on itself
 - There is a $3 > 1$ rule that can be described using 8 bits
 - Therefore there are 256 possible rules
 - Which are interesting?
- Chocolate bar problem
 - $n \text{ pieces} = a \text{ cuts} + 1$
- Various expressions for pi
 - $1/1 - 1/3 + 1/5 - 1/7 \dots$
 - Average of flips until heads exceeds tails (SM)
 - Dart board unit square / unit circle
 - Speed up by not needing to use square root (see time function)
- Five foot copper pipe
 - how to work in a magnet?
 - extrapolation of dimension... what size for a 100-dimensional box?
- What is the Name of this Book?

- Lion and Unicorn
 - Lion: Yesterday was one of my lying days. Unicorn: Me too!
 - Lion: I lied yesterday. I will lie again 2 days after tomorrow.
 - Lion: I lied yesterday. I will lie again tomorrow. (nope)
 - Lion: I lied yesterday and I will lie again tomorrow. (yep)
- Knights and Knaves
- Knights
 - The Tour (and Reentrant)
 - The Trapped Knight (spiral board square numbering)
- Fibonacci
 - Generate the series
 - Generate the ratios: To what do they converge?
 - Generate the spiral
 - Generate Meru Prastarah and produce Fibonacci
 - Back-propagate MP + F
 - Even-odd mask of MP is the SG
 - Also matches the XOR Universe
 - What about a 2D CA? Life? 2^9 possible rules (1/2 million)
- L-Systems
 - x, y, +, -
 - Suggestion: Start with an xy+- string and write the render turtle first
 - This gets to an intermediate punchline directly
 - Then we have a landing zone for the $x > y + x + y$; $y > x - y - x$ recursion
- David Hilbert's hotel
 - Strangeness of infinity
 - Pi knows where you left your house key
- The circular ring of numbers problem
- Proofs book: 2^{n-1} L-Domino tiling
 - to the nature of induction proof
- Four bugs
- Gargutron
- Eye color
- Mobius derision
- Planar graphs and Platonic solids

Poetry and Philosophy

- Ladle Rat
- Twas brillig
- Strogatz on loving the problem
- "Gardnerian" as criteria for worthy problems
- Why do this?
 - Remind... what is it Sagan had to say?
 - Satisfactory philosophy of ignorance
- Let's talk about AI for a moment
 - equivalence to "your mom does your homework"
- Work versus Distraction
- The Unreliable Narrator and the social contract of instruction

The L-System Code-Along

```
from turtle import Turtle, done, tracer, update, Screen
screen = Screen()
screen.setup(width=700, height=700)

def CreateDirections(s, n):
    # Simon Says: return "+-+-+-+---+-"
    for i in range(n):
        s1 = ""
        for c in s:
            if c == "x":    s1 += "y+x+y"
            elif c == "y": s1 += "x-y-x"
            else:          s1 += c
        s = s1
    return s

go    = 9
n     = 6
bend  = 60
d = CreateDirections("x", n)
print(d)

t = Turtle()
t.pencolor("blue"); t.setheading(60); t.penup()
if n % 2: t.goto(0, -300)
else: t.goto(-300, -300)
t.pendown()
tracer(0, 0)

for c in d:
    if c == "+": t.left(bend); t.forward(go)
    elif c == "-": t.right(bend); t.forward(go)

update()
done()
```